

	According to definition for thermodynamic temperature scale.
12	D As the temperature has stabilised, the rate of increase of internal energy $(\Delta U) = 0$.

	The filament is hotter than its surroundings, loses heat (Q). Thus, rate of heating the filament is negative. Positive work (W) is done on the filament by electric current (recall the potential difference across a component in a circuit as the work done to drive a unit charge through the component).
13	A